10.4.2 Controlling network access to an RDS database

An RDS database is linked to security groups. Each security group consists of rules for a firewall controlling inbound and outbound database traffic. You already know about using security groups in combination with virtual machines.

The next listing shows the configuration of the security group attached to the RDS database in our WordPress example. Inbound connections to port 3306 (the default port for MySQL) are allowed only from virtual machines linked to the security group called `WebServerSecurityGroup`.

Listing 10.6 CloudFormation template extract: Firewall rules for an RDS database

```
DatabaseSecurityGroup:                                    ①
  Type: 'AWS::EC2::SecurityGroup'
  Properties:
    GroupDescription: 'awsinaction-db-sg'
    VpcId: !Ref VPC
    SecurityGroupIngress:
    - IpProtocol: tcp
      FromPort: 3306                                      ②
      ToPort: 3306
      SourceSecurityGroupId: !Ref WebServerSecurityGroup  ③
```

① The security group for the database instance, allowing incoming traffic on the MySQL default port for web servers

② The default MySQL port is 3306.

③ References the security group for web servers

Only machines that really need to connect to the RDS database should be allowed to do so on the network level, such as EC2 instances running your web server or application server. See chapter